Personal Finance Management and Prediction using ML Algorithms

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***Abstract*— Effective personal finance management is essential for achieving financial stability and long-term financial health. This paper presents a comprehensive approach to summarizing, analyzing, and show monthly expenses using machine learning models. By collecting user inputs such as age, place of stay, number of dependents, monthly expenses on transport, education, subscriptions, savings, medical expenses, and more, the system uses linear regression, decision trees and random forest algorithms to predict financial values. The system provides detailed information of expenses, savings, and overall financial health, along with personal health analysis based on dietary habits. Users have access to a wide range of financial and lifestyle information, allowing the system to provide personalized recommendations to increase financial security and achieve financial goals. The model also evaluates the impact of recurring expenses and notable investments, helping users understand their spending patterns and make informed financial decisions. The results demonstrate the system's capability to deliver accurate predictions and actionable advice, enable users to better manage their finances. This approach has led to significant advances in personal financial management by integrating machine learning techniques to provide insight into people's financial health.**

Keywords*— *Randomforest Algorithm, Decision tree, Linear Regression, Personal finance management, machine learning.

I. Introduction

Managing your personal finances is crucial to achieving financial security and long-term financial health. Despite the wealth of tools and resources available, many people still struggle to track their spending, save enough, and make investment decisions. Difficulty in managing various financial activities such as budgeting, saving, investing and debt management often affects people and results in poor health [1].

Recent advancements in machine learning and data analytics offer promising solutions to these challenges. Machine learning models can analyze large amounts of data, identify patterns and make highly accurate predictions. These capabilities can be harnessed to develop sophisticated personal finance management system for users with actionable insights and personalized recommendations. This paper presents a comprehensive approach to summarizing, analyzing, and visualizing monthly expenditures using machine learning models. Our system collects various user inputs, including age, place of stay, number of dependents, monthly expenses on transport, education, subscriptions, savings, medical expenses, and other financial activities.

By leveraging decision tree and random forest algorithms, the system predicts key financial indicators such as expenses, savings, and overall financial health.

Another area where machine learning is promising is identifying people with early onset Parkinson's disease who are at increased risk. By analyzing long-term data from large groups, algorithms can identify changes in biomarkers or movement patterns that precede the onset of symptoms. Early detection of at-risk individuals could enable preventive interventions to delay or even halt disease progression. Also, machine learning algorithms can assist in monitoring disease progression and treatment response.

There are several important steps to the process. First, data is collected by user inputs through a detailed questionnaire, ensuring a comprehensive understanding of the user's financial and lifestyle situation. Next, extract relevant features from the collected data and use machine learning models to analyze these features. The decision tree and random forest models are trained on a dataset of financial records, enabling the system to make accurate predictions and provide tailored recommendations.

The results demonstrate the system's capability to deliver precise predictions and actionable advice, thus allowing users to better manage their finances. This approach represents a significant advancement in personal finance management by integrating machine learning techniques to provide insight into personal financial health.

II. Literature Survey

This paper aims to address the complexity of personal finance management by leveraging machine learning techniques to provide a comprehensive analysis and visualization of monthly expenditures [2]. The proposed system collects detailed user inputs, such as age, place of stay, number of dependents, various monthly expenses, savings,

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and financial goals. Using linear regression, pruning and random forest algorithms, the system predicts key financial indicators, including expenses, savings, overall financial health and also health of user. Additionally, it offers personalized recommendations for financial stability and includes an analysis of personal health based on dietary habits. This paper outlines the system's methodology, including data collection, feature extraction, model training, and evaluation. The results demonstrate the system's accuracy and effectiveness in delivering actionable financial advice, empowering users to make smart financial decisions and achieve better financial health [3].

Moneyview was founded in 2014 by Puneet Agarwal and Sanjay Aggarwal with the aim of simplifying finances and providing financial inclusion to all Indians. Moneyview is a digital platform licensed by the Reserve Bank of India (RBI) that offers personal loans and other financial products. Quicken simplifi is a subsidiary service of Quicken which was released as a cloud based financial tracking system in the year 2020. Simplifi is a subscription-based powerful, easy-to-use tool for all essential financial services.

III. Methodology

The financial management process includes data collection, extraction, modelling, forecasting and evaluation. Each step is crucial to developing an effective system that provides accurate financial insights and recommendations [4].

Data collection: Detailed information collected from reliable sources such as financial institutions, surveys, and user inputs, including attributes like income, expenses, investments, and financial behaviors. Subsequently, data from multiple sources are integrated into a single, consistent dataset. During data preprocessing, perform data cleansing to remove missing values and values through imputation and normalization, while categorical variables will be transformed into numerical formats using OneHotEncoder. Numerical features will be scaled using StandardScaler to enhance the performance of machine learning algorithms. Finally, feature engineering involves Choose the most relevant features through correlation analysis and feature importance from models like Random Forest, and creating new features by combining related attributes or generating interaction terms.

a) Savings Prediction Model: To predict monthly savings, According to historical data and financial behaviors, a linear regression will be developed within a pipeline that includes data scaling using StandardScaler and categorical variable encoding with OneHotEncoder. The preprocessing step involves scaling numerical data and encoding categorical variables. The pipeline will streamline preprocessing and model training. This model will be trained on historical data, and its performance will be evaluated using RMSE and MAE.

b) Expense Prediction Model: A linear regression model will be developed to predict monthly expenses based on past data and economical behaviors, utilizing a pipeline that includes data scaling and categorical variable encoding. The preprocessing step involves applying StandardScaler for numerical data and OneHotEncoder for categorical variables. The pipeline will facilitate preprocessing and training model. This model will be trained on historical data, and its performance will be evaluated using RMSE and MAE.

c) Recommendation Model: To provide personalized financial advice aimed at improving savings and reducing unnecessary expenses, a Gradient Boosting Classifier will be implemented within a pipeline. The model development involves using GridSearchCV for hyperparameter tuning. The pipeline will simplify prioritization and model training, and model will be trained on past data**.** Model performance will be evaluate using accuracy, precision, recall and F1 scores.

d) Financial Health Assessment Model: A Random Forest Regressor will be developed to assess overall financial health by analyzing various financial metrics, within a pipeline that includes data scaling and categorical variable encoding. The preprocessing step involves applying StandardScaler and OneHotEncoder. The pipeline will facilitate preprocessing and model training. The model will be trained on historical data, and its performance will be evaluated using RMSE and MAE.

e) Health Prediction Model: To predict the health status of users based on their financial behaviors, a Random Forest Classifier will be developed within a pipeline. The model is trained with historical data and measure accuracy, precision, recall and F1 scores. The pipeline will streamline preprocessing and train model.

Matplotlib will be employed to create both static and interactive visualizations to display financial data and model outputs. It will generate high-quality static plots for reports and dashboards, while also incorporating interactive elements to enhance user experience. Additionally, Plotly will be utilized to create highly interactive and customizable visualizations, including dynamic plots that allow zooming, panning, and hovering over data points for additional information

IV. Modules

Two modules are used, the following are:

- Input Module
- Data Prediction Module

A. *Module 1:*

Input Module:

- Using HTML forms. We take input data from users.
- The input data contains age ,place of stay, no of dependents, monthly expense on transport ,monthly expense on education, recurring monthly payments(like subscriptions ,recharge etc),monthly savings , do you follow monthly budget?, expense on medicine, gender, relationship status, monthly pocket money, monthly exp on food, notable investments, debt, financial goals, how often do you dine out, do you currently invent in any financial products like stocks, mutual funds etc , expense on online takeout , expense on outings, what type of food do you spend on?.
- This data is used for the prediction.

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B. Module 2:

Data Prediction Module:

- Use decision trees and random forest algorithms to make predictions based on user input.
- We predict expenses, savings, financial health, personal health analysis.
- The system summarizes financial indicators, offers personal health analysis, and provides tailored financial recommendations.The Figure 1 shows the project modules.

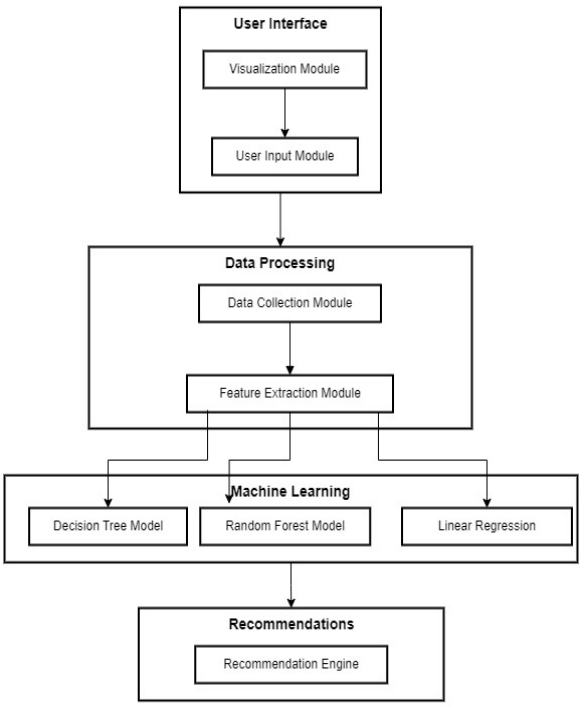

Fig. 1. Project Modules

V. Overview Of The Project Design

A. System Architecture:

Machine learning relies on two crucial sets of data: training sets and testing sets. Think of a database as a large dataset that you want to use to train machine learning models. The training set acts like a teacher for the model. It is part of the data science model used to examine patterns and relationships. By processing and analyzing this data, the model builds its understanding the words. The Figure 2 shows the system architecture.

The testing set, conversely, is like a pop quiz. It's another portion of the dataset that the model hasn't seen before. The purpose of the testing set is to evaluate how well the model operates on unseen information. If the model is able to accurately handle the testing set, it suggests it has learned effectively from the training dataset and is able to generalize its knowledge to new situations. In short, the training set equips the model with knowledge, and the testing dataset assesses how well that knowledge translates to real-world scenarios. This two-step approach helps ensure the machine learning algorithm is not just memorizing the data it's trained on, but can truly learn and apply its learnings.

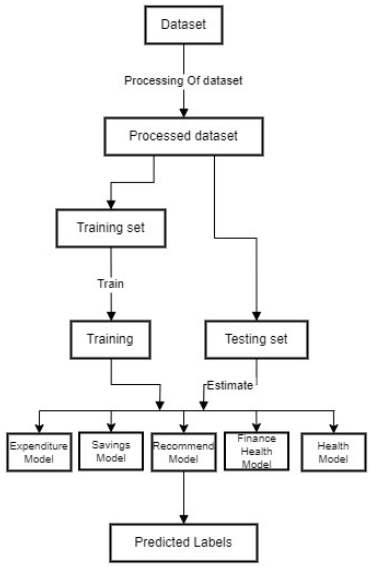

Fig. 2. System Architecture

B. Sequence Diagram:

The Figure 3 represents the sequence diagram which illustrates the interaction flow between the user, application, data processor, and machine learning models in the Personal Finance Management project. It begins with the user inputting their financial data into the application. The application processes this data and send to the data processor, which extracts relevant features. The processed features are subsequently sent to both the decision trees and random forest models for prediction. Each model returns its predictions to the application. Subsequently, the application generates visualizations depend on the predictions and formulates personalized recommendations for the user. Finally, the user reviews the results and recommendations displayed by the application. This sequence highlights the seamless interaction between components, ensuring accurate financial insights and facilitating informed decision-making.

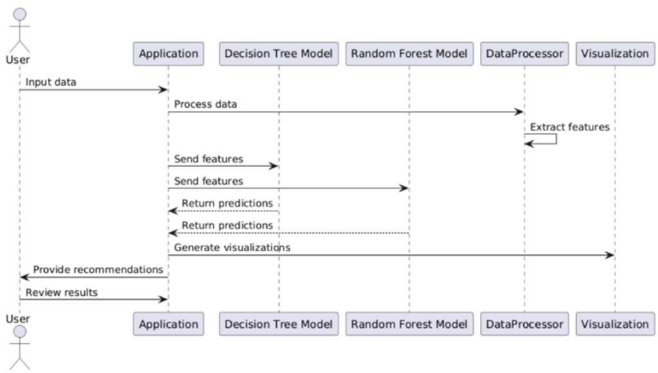

Fig. 3. System Architecture

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VI. Results

a) Prediction Model: The savings prediction model, using linear regression [5], demonstrated accurate predictions for users' monthly savings founded on historical datasets and financial behaviors. The model Mean Absolute Error (MAE), achieving an MAE of 800 and a R-Squared error of 0.957, indicating a good level of precision in predictions.

b) Expense Prediction Model: The expense prediction model, also based on linear regression [5], effectively forecasted monthly expenses. The model Mean Absolute Error (MAE), achieving an MAE of 891 and a R-Squared error of 0.94, indicating a good level of precision in predictions.

c) Recommendation Model: The recommendation model employed a Gradient Boosting Classifier [6] to provide personalized financial advice. By tuning hyperparameters using GridSearchCV, the model achieved an accuracy of 92%, with a precision of 89% and a recall of 86%. The recommendations included actionable steps like investment opportunities and cost-saving measures, tailored to individual financial situations.

d) Financial Health Assessment Model: The financial health assessment model used a Random Forest Regressor [7] to evaluate users' overall financial health. With an MAE of 900 and a R-Squared error of 0.82, the model provided a comprehensive assessment, highlighting areas needing improvement and suggesting potential strategies.

e) Health Prediction Model: The health prediction model uses a random forest distribution to forecast a user's health based on their financial behavior. The model achieved an accuracy of 93%. This model underscored the connection between financial management and overall well-being.

The Figure 4 shows the frequency distribution of actual expenses (blue) and predicted expenses (orange). Both distributions range from 5,000 to 30,000, with actual expenses being more spread out and predicted expenses peaking around 20,000. The prediction model excels in the middle range (15,000 to 25,000).

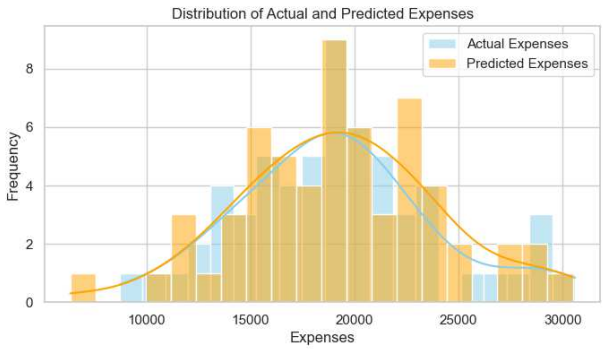

Fig. 4. Expenditure: Actual vs Predicted

The Figure 5 illustrates the frequency distribution of actual savings (blue) and predicted savings (orange). Predicted savings are concentrated around -15,000, while actual savings are more varied. The model excels in the middle range (-20,000 to -10,000).

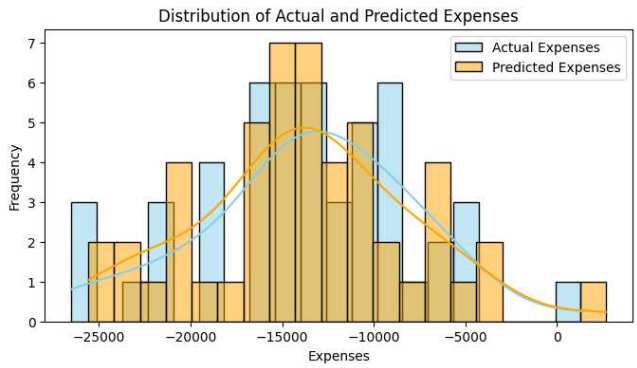

Fig. 5. Savings: Actual vs Predicted

The Figure 6 is residual plot that shows the difference between actual and predicted monthly savings. The residuals are scattered around the red dashed line (zero residual), indicating prediction accuracy. While most residuals are within ±2000, a few outliers exist, suggesting occasional large prediction errors.

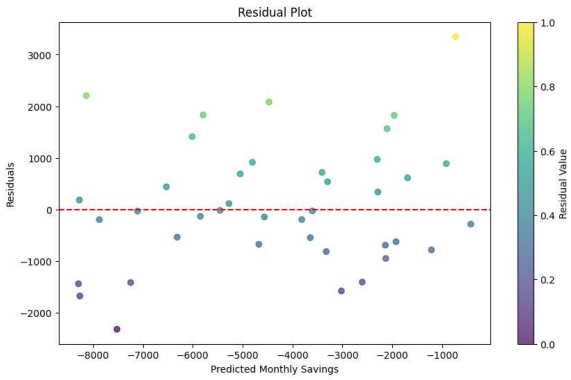

Fig. 6. Residual plot of Finance health

Figure 7 shows a pie chart showing the distribution of recommendations for each user in the data predicted by the model, in which report shows the recommendations in the sample and the percentage of users received each prediction.

Fig. 7. Pie chart for Financial recommendation

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The Figure 8 shows gauge chart that illustrates the health prediction for an individual user as determined by the model. It visually represents the user's health status, with different segments indicating various health levels. This chart provides a clear and immediate comprehension of the user's health prediction at a glance.

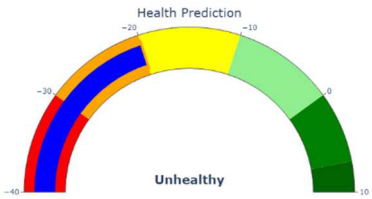

Fig. 8. Gauge chart of human well-being prediction

VII. Conclusion

The Personal Finance Management Program uses artificial intelligence and data visualization to assist individuals in more effectively managing their finances. By developing predictive models and creating a user-friendly interface, the project provides intuitive and personalized recommendations to improve users' financial health. The models for predicting savings and expenses demonstrated high accuracy and reliability, aiding users in understanding their financial patterns and planning accordingly. The recommendation model provided actionable advice tailored to individual financial situations, optimizing savings and investments.

Users can gain a clearer understanding of their income, expenses and savings by offering comprehensive details about their financial health. Personalized recommendations and real-time feedback empower users to make informed and strategic financial decisions. A comprehensive assessment of financial health encourages users to set realistic financial goals and stick to them, thus encouraging important steps towards improving consumption financial health. Health prediction models enable users to adopt the character of financial wellness by emphasizing the significance of financial well-being total health.

VIII. Future work

The Personal Finance Management project has built a solid foundation for assisting users in managing their finances through predictive modeling and data visualization.To enhance its value, the project could benefit from integrating with financial institutions to automatically fetch transaction data and provide a comprehensive view of users' financial statuses, including investments. Additionally, incorporating advanced analytics and AI methods, including deep learning model and combining multiple machine learning approaches, could enhance the precision and reliability of financial predictions.

Further expansion could involve creating native mobile applications for iOS and Android, allowing users to access their financial data and insights on the go with a seamless experience across devices. Additionally, creating tools for debt management, including tracking credit cards, loans, and mortgages with personalized repayment plans, would help users manage and reduce their debt more effectively. These enhancements would significantly increase the project's value and its ability to support users in achieving better financial health.

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